MAD Fall 2026 Project 1

Project for Mobile Application Development

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Status: Undergraduates
Project Topic: Campus Challenge
Semester: Fall 2026
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Section 1 - Intro:

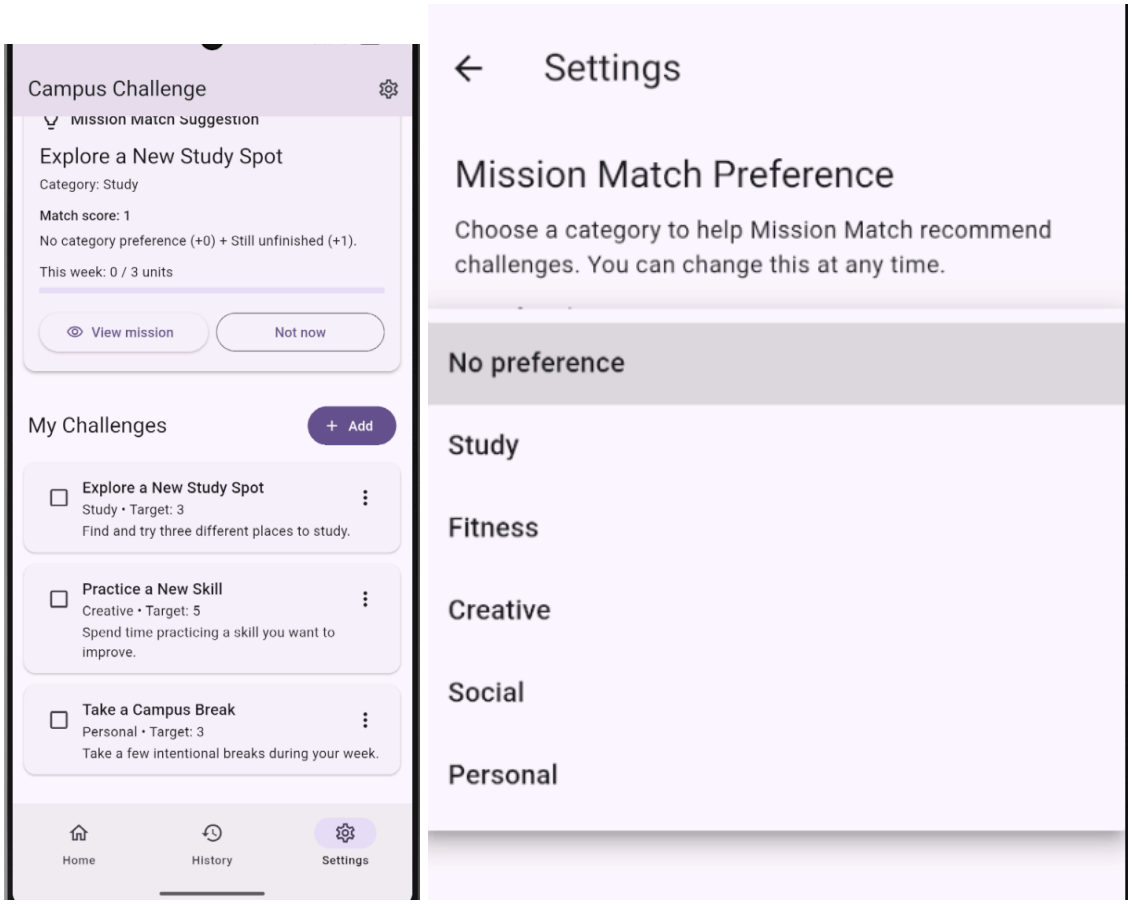
For this project, my partner and I were tasked with creating an app based on a real world scenario we care about. The topic we chose was campus challenge micro Adventures because it was something that related to us due to being in a large commuter campus as upperclassmen, but having trouble making time to use it to our full advantage. This app is a personal Dashboard for users to mark achievements and challenges for themselves as they pursue their degree.

Section 2 - Core:

The Personal Campus Challenges app has five main screens, which are the home screen, create and edit challenge form, challenger detail screen, history and insights screen, and settings screen. All of these screens work together to provide students with a simple interface to track and create challenges to get better acclimated to their environment. Upon opening the app, the first screen that the user sees is the home page, which displays the mission card AI helper as well as all the challenges that the user has. From this page you can do everything, including add challenges, navigate to different screens, and modify the challenges you have. As you move through the app, you will move to a form that prompts you to create a challenge.

On this form you can interact with multiple text boxes to give value to your challenge, name it, and include activities or checklist items for it. As it is utilized, validation helps prevent invalid input like blank text boxes and unsaved inputs. This is important not only for the database to hold valid information but also for the user to be able to look back and understand the inputs for every challenge and activity. Upon submission of the challenge, the user can navigate the challenge details screen, which will show all information about the challenge, whether it was just created or created in the past. This allows the user to see descriptions for the challenge and keep up with their progress for the task, since they have the ability to log units and notes.

If the user ever wanted to delete a note, they can delete on this screen, whereas if they wanted to delete a challenge, they would delete it on the home screen. As more and more information is being logged for each challenge, the user can then move to the history and insights page that will allow the user to review all of their past activity and challenges by displaying a comprehensive list of logged activity and completed activity. If the user wants to make modifications to the app/AI helper based on their personalized needs, they can go to the settings screen, which will allow them to modify and save preferences they have, even if they are offline



Section 3 - Database:

Our app uses SQLite for data persistence with items related to logging, activity changes, and challenge creation. The SharedPreferences are for user settings. To keep everything structured our app has a dedicated class called DatabaseHelper that manages the local database connection, tables, and CRUD operations.

Section 4 - Smart Helper:

Mission Match uses information already stored in the application. It considers the user's saved preferred category and the amount of progress recorded for each challenge during the current week.

The helper assigns points using two rules:

Category match: +2 points. A challenge receives two points when its category matches the user's saved preference.

Unfinished weekly target: +1 point. A challenge receives one point when its target has not yet been met during the current week.

After calculating the scores, the helper ranks challenges from highest to lowest. If two challenges have the same score, the challenge with the lower ID is ranked first.

The helper uses the current local calendar week, beginning Monday at midnight and ending the following Monday at midnight, when calculating weekly progress.

Baseline and User Control

The helper compares its ranked recommendation against a simple baseline: the first valid challenge with the lowest ID. This provides a basic comparison to show whether the scoring rules change which challenge is recommended.

The recommendation card displays the suggested challenge, its score, and an explanation of why it was selected. Users can select View Mission to open the challenge or Not Now to dismiss the recommendation.

The helper does not automatically create, start, or complete challenges. It only provides a recommendation, leaving the final decision to the student.

If there are no challenges, the helper displays a message asking the user to add one. If the available challenges are invalid, it displays a message explaining that no valid missions are available.

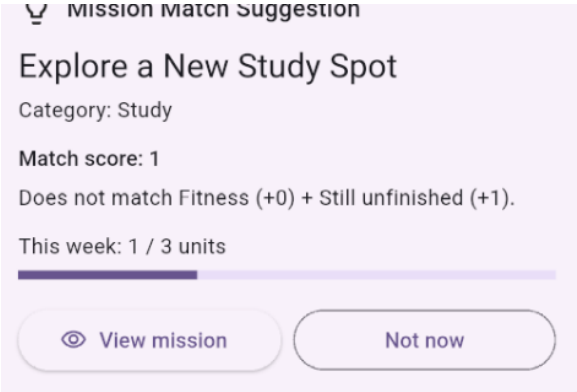


Section 5 - Testing:

To help us ensure that the create update and delete operations were working while we did three test cases for the challenge aspect of the app. The first test was to test creating a new challenge and seeing how it backed filled as well as showed on the Home screen. The expected result was that the challenge would show on the home screen as well as show a success message This did not happen during our initial test because we did not complete the form output correctly for it to not only show with the other challenges in the widget, but to show in the SQL as well. This was addressed and debugged along with other database issues.

For the admission match helper We also did a series of tests. The first test we did was test its normal Outcomes by utilizing the app like a typical user would. With this test the mission match ran as expected, showing helpful suggestions and logic. The next case we did included no data, where the helper was tested with no history. The helper In this case showed an error message saying there was not enough data to assist.

An additional test we ran included putting out of range information into the app. We did this by keeping user preferred choices and their activity blank. Next we tested the counter example case where Ai helper should not be giving suggestions. This would happen based on the user's activity and their placement within the app. To finish up testing, we tested the boundary of the helper by creating two challenges that are different but gave the user the same match score. The goal is for the helper to be able to choose the mission with the lowest ID versus trying to pick off of point value, which would as expected.



Section 6:

Future Enhancements & Reflection